

NETWORKS AND COMPLEXITY

Solution 13-9

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 13.9: The Zombie Apocalypse ... will be weird [Lvl. 3]

During the zombie apocalypse, each zombie is killed within a day, but not before infecting 3 people with the zombie virus. Within a day, the infected people turn to zombies.

- a) Model the system as a discrete-time map where the Z_t and I_t are the numbers of zombies and infected on day t .

Solution

Using the information from the question, we can model the system by the discrete-time map

$$Z_{t+1} = I_t \tag{1}$$

$$I_{t+1} = 3Z_t \tag{2}$$

- b) Use the method from the chapter to compute Z_t , I_t for all $t > 0$.

Solution

We define the vector

$$\mathbf{X}_t = \begin{pmatrix} Z_t \\ I_t \end{pmatrix}. \tag{3}$$

We can now write the system in matrix form

$$\mathbf{X}_{t+1} = \mathbf{J}\mathbf{X}_t, \tag{4}$$

where

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix}. \tag{5}$$

To compute the eigenvectors of $\mathbf{J}$ we solve

$$\lambda \begin{pmatrix} 1 \\ x \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ x \end{pmatrix} \tag{6}$$

which we can also write as

$$\lambda = x \tag{7}$$

$$\lambda x = 3 \tag{8}$$

Substituting the first line into the second one gives us

$$x^2 = 3 \tag{9}$$

and hence

$$x = \pm\sqrt{3}. \quad (10)$$

Using $\lambda = x$ we have now two eigenvalues

$$\lambda_1 = \sqrt{3}, \quad \lambda_2 = -\sqrt{3} \quad (11)$$

and the corresponding eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \quad (12)$$

Therefore the general solution is

$$\begin{pmatrix} Z_t \\ I_t \end{pmatrix} = c_1(\sqrt{3})^t \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} + c_2(-\sqrt{3})^t \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \quad (13)$$

- c) Using your solution, consider the case where the outbreak starts with one infected on day $t = 0$ and compute the number of zombies on days 1, 2, 20, and 21.

Solution

To find the particular solution we need to determine c_1 and c_2 by solving the system on day 0,

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \quad (14)$$

The upper line reads

$$0 = c_1 + c_2, \quad (15)$$

hence,

$$c_1 = -c_2. \quad (16)$$

Substituting this into the lower line yields

$$1 = c_1\sqrt{3} + c_1\sqrt{3} = 2c_1\sqrt{3}. \quad (17)$$

Hence,

$$c_1 = \frac{1}{2\sqrt{3}} \quad c_2 = -\frac{1}{2\sqrt{3}} \quad (18)$$

We now have the particular solution

$$\begin{pmatrix} Z_t \\ I_t \end{pmatrix} = \frac{1}{2\sqrt{3}}(\sqrt{3})^t \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} - \frac{1}{2\sqrt{3}}(-\sqrt{3})^t \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \quad (19)$$

We can write the line for the Zombies as

$$Z_t = \frac{1}{2\sqrt{3}}(\sqrt{3})^t - \frac{1}{2\sqrt{3}}(-\sqrt{3})^t. \quad (20)$$

By pulling a factor $(-1)^t$ from the second term, we can further simplify this to

$$Z_t = \frac{1}{2\sqrt{3}} \left((\sqrt{3})^t - (-1)^t(\sqrt{3})^t \right). \quad (21)$$

And by canceling a factor of $\sqrt{3}$ and collecting the terms, we arrive at

$$Z_t = \frac{1 - (-1)^t}{2}(\sqrt{3})^{t-1}. \quad (22)$$

From this form, it is now apparent that the number of zombies will be 0 on all even days. On odd days, the first factor becomes 1, and we have

$$Z_t = (\sqrt{3})^{t-1} \quad [\text{On odd days}] \quad (23)$$

Hence the number of zombies is zero on day 2 and day 20. On day 1 it is $\sqrt{3}^0 = 1$ and on day 21 it is $\sqrt{3}^{20} = 3^{10}$ which is 59049. Of course it will be zero again on day 22.